



National University
of computer and emerging sciences

Computer Organization and Assembly Language (EL-2003)

Semester: Fall 2025

Section: BCS-3D / BCS-3H

Course Instructor: Muhammad Owais

LAB # 01

Getting Started with Assembly Programming Language (x86) in Visual Studio 2022 using Irvine Library

Lab Objectives:

By the end of this lab, students will be able to:

1. Understand the core concepts of Assembly Language and its role in computer systems.
2. Recognize the importance and applications of Assembly in various domains.
3. Successfully install and configure Visual Studio 2022 with the Irvine Library for MASM programming.
4. Create, assemble, link, and execute a basic Assembly program.
5. Apply step-by-step setup techniques to ensure a smooth programming environment.
6. Troubleshoot and resolve common setup, compilation, and linking errors.
7. Utilize Visual Studio's debugging tools, including setting breakpoints, stepping through code, and observing register/memory states during execution.

Introduction to Assembly Programming Language

Assembly language is a type of programming language that is designed to be used by developers to write programs that can run directly on a computer's central processing unit (CPU). It is a **low-level language**, which means it is closer to the machine code the CPU can execute, making it more powerful than other higher-level languages such as C++, Java, or Python.

In an assembly language program, each instruction represents a single operation that the computer's CPU can perform. These can include simple arithmetic and logical operations, such as adding and subtracting values, as well as more complex operations that involve manipulating data stored in the computer's memory. Assembly language programs are typically written in a text editor and then assembled using a specialized software tool called an **assembler**.

Importance of Assembly Programming Language

1. Close to Hardware (Low-Level Control)

Assembly language gives programmers direct access to a computer's hardware. Unlike high-level languages, it allows you to directly manipulate CPU registers, memory addresses, and I/O ports. This is essential when developing [device drivers](#), [firmware](#), or [embedded systems](#), where you need precise control over the hardware.

2. High Performance and Efficiency

Since assembly instructions are closely mapped to the CPU's native machine code, programs written in assembly run [faster](#) and are more [memory-efficient](#). It is often used in situations where every CPU cycle matters, such as [real-time systems](#), [gaming engines](#), or [robotics](#).

3. Essential for Embedded Systems

Many embedded devices (like microcontrollers in washing machines, cars, or IoT devices) have very limited processing power and memory. Assembly is often used to write [small](#), [fast](#), and [highly optimized programs](#) for these devices.

4. Better Understanding of Computer Architecture

Learning assembly helps you understand how the CPU, memory, and other components interact. You gain practical knowledge about [instruction execution](#), [registers](#), [stack operations](#), and [memory addressing](#), which is also useful for higher-level programming and debugging.

5. Optimizing High-Level Code

Even if a program is written in a high-level language like C++ or Python, certain performance-critical sections can be optimized using inline assembly. This helps speed up tasks like [graphics rendering](#), [encryption](#), or [scientific computation](#).

How Does Assembly Work?

The syntax of assembly language varies depending on the specific machine architecture it is being used with. However, most assembly languages share a few basic features. Assembly language programs are typically made up of a series of instructions written using a combination of mnemonic codes and operands, representing the data being manipulated by the instruction.

Let's understand how assembly language works in detail.

How Assembly Language Works

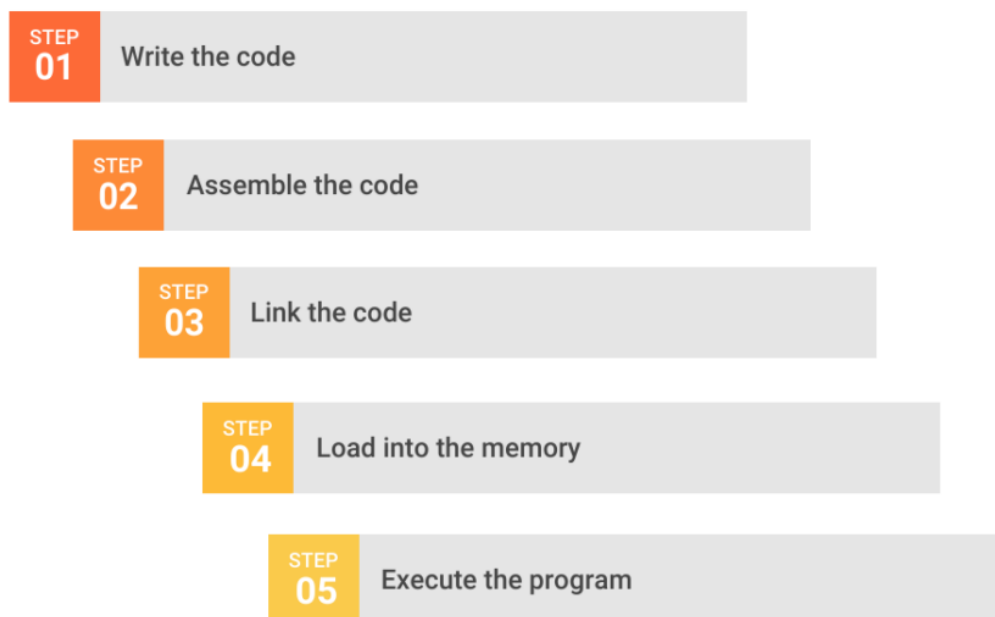


Figure 1: How Assembly Language Works

Step 1: Writing the code

The first step is to write the code in assembly language. Assembly language code consists of mnemonic instructions that correspond directly to the machine language instructions executed by the CPU.

For example, here is a code in x86 assembly language that adds two numbers:

```
mov eax, 5 ; move the value 5 into the ebx register  
mov ebx, 7 ; move the value 7 into the ebx register  
add eax, ebx ; add the values in eax and ebx and store the result in eax
```

In this code, the first two lines set the values of the eax and ebx registers. The third line adds the values in the eax and ebx registers and stores the result in the eax register.

Step 2: Assembling the code

The next step is to assemble the code using an assembler. An assembler is a program that converts the assembly language code into machine language the computer can execute. The assembler reads the assembly code and translates it into binary machine code, a series of 0s and 1s representing the instructions and data in the program.

When the assembler converts the assembly code into machine code, it generates a file with an .obj extension containing the machine code and other information that the linker uses to create the final executable file.

Step 3: Linking the code

The next step is to link the code. Linking is the process of combining the object file generated by the assembler with any necessary system libraries to create an executable program. During the linking process, the linker resolves any external references to functions or variables and combines all the object files into a single executable file.

Step 4: Load into the memory

Once the code has been linked, it needs to be loaded into memory. The operating system accomplishes this. The executable file is loaded into a specific location in memory, and the operating system sets up the program's environment, including its stack, heap, and global variables.

Step 5: Executing the program

The final step is to execute the program. When the CPU reads the instructions, it will execute them one by one. The machine code instructions generated by the assembler correspond directly to the assembly language instructions in the original code.

Visual Studio

Visual Studio is an **integrated development environment (IDE)** created by Microsoft. It's basically a powerful software application that provides all the tools you need to write, edit, debug, and manage code for different types of applications.

For this course, we will use Visual Studio version 2022 to develop programs in Assembly Language. We could, however, use a stand-alone assembler like NASM or MASM to code in Assembly Language.

Steps to Install Visual Studio 2022

Step 1: Go to the Download Page

1. Open your web browser and go to: <https://visualstudio.microsoft.com/downloads/>
2. Scroll down to find **Visual Studio 2022**
3. Click Download for the **Community Version**.

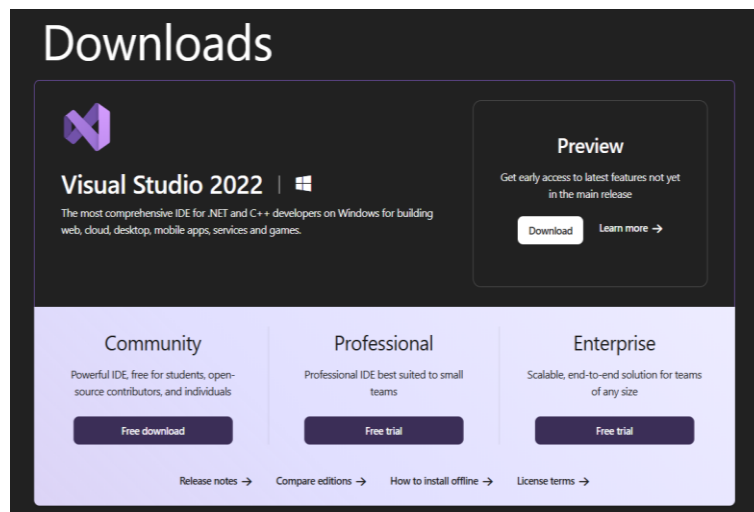


Figure 2: Downloading Page - Visual Studio

Step 2: Download the Bootstrapper File

1. Once you click [Download](#), a small setup file (bootstrapper) will be downloaded.
2. The file name will be something like:
 - VisualStudioSetup.exe
 - OR vs_community.exe

Step 3: Initiate the Installation

1. Go to your [Downloads](#) folder.
2. Double-click the downloaded bootstrapper file.
3. If prompted by [User Account Control](#), click [Yes](#).
4. Read the [Microsoft License Terms](#) and the [Privacy Statement](#), then click [Continue](#).

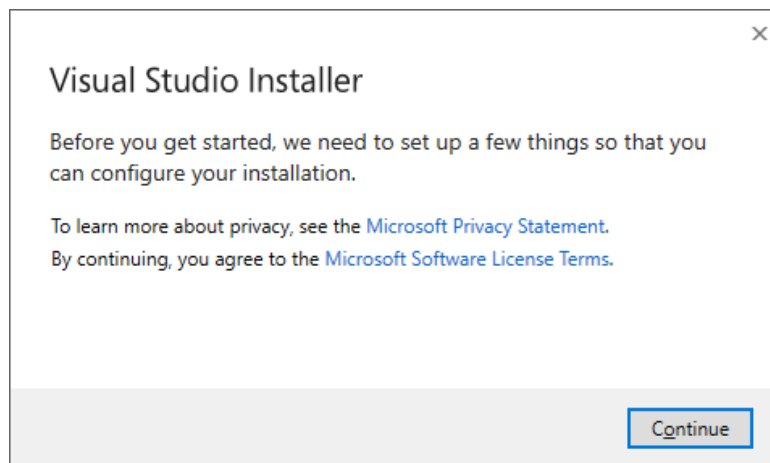


Figure 3: Visual Studio Installer

Step 4: Choose Workloads

1. The [Visual Studio Installer](#) will open.
2. From the list of workloads, select [Desktop development with C++](#) (mandatory for C++ development).
3. Review workload summaries to confirm the features you need.
4. After selecting workloads, click [Install](#).

Step 5: Installation Progress

1. You will see progress bars showing the installation status.
2. Wait until the process completes; this may take several minutes depending on your internet speed and selected workloads.

Step 6: Launch Visual Studio 2022

1. Once the installation is complete, click [Launch](#) from the installer.
2. Sign in with your Microsoft account (optional, but recommended).
3. Choose your preferred development settings and theme.

Visual Studio 2022 is now ready to use!

Introduction to the Irvine Library

The [Irvine Library](#) (formally called *Irvine32.inc*) is a collection of procedures, macros, and definitions created by [Kip Irvine](#) to make it easier for students and beginners to learn [Assembly Language Programming](#) (specifically x86 Assembly on Windows). It comes with the textbook *Assembly Language for x86 Processors* by Kip Irvine.

The library provides functions for:

- Displaying strings and numbers
- Reading input from the keyboard
- File handling
- Random number generation
- Simple screen formatting

Why Do We Use It?

1. [Simplifies Learning](#)

Assembly is very low-level, and even printing "Hello World" requires many lines of code (using system calls or interrupts). Irvine library makes this simple with easy-to-use procedures. Example:

```
INCLUDE Irvine32.inc

.code
main PROC
    mov edx, OFFSET msg ; point to string
    call WriteString     ; print string
    exit
main ENDP

.data
msg BYTE "Hello, World!", 0
END main
```

Without the library, you'd need to use int 21h (DOS) or Windows API calls, which are much harder for beginners.

2. Focus on Concepts

Instead of spending too much time on low-level OS calls, students can concentrate on [assembly instructions, registers, stack, loops, and procedures](#).

3. Cross-Platform Learning (within Windows)

It hides OS-specific details (like system calls), so the same simple code runs on any Windows system with MASM + Irvine library.

[Click here to download Irvine library from this](http://www.asmirvine.com/gettingStartedVS2019/Irvine.zip)

link: www.asmirvine.com/gettingStartedVS2019/Irvine.zip

Once you have downloaded the required Irvine library, install it in your computer and verify that a folder named Irvine has been created in your C:\ drive.

Configuring Irvine32 Library with Visual Studio

Follow the steps below to create and configure a Visual Studio project for Assembly Language using the [Irvine Library](#):

Step 1: Create a New Project

1. Open [Visual Studio 2022](#).
2. Click on [Create a new project](#).
3. In the [template search bar](#), type [Empty Project](#) and select it.
4. Click [Next](#), give your project a [name](#), and click [Create](#).

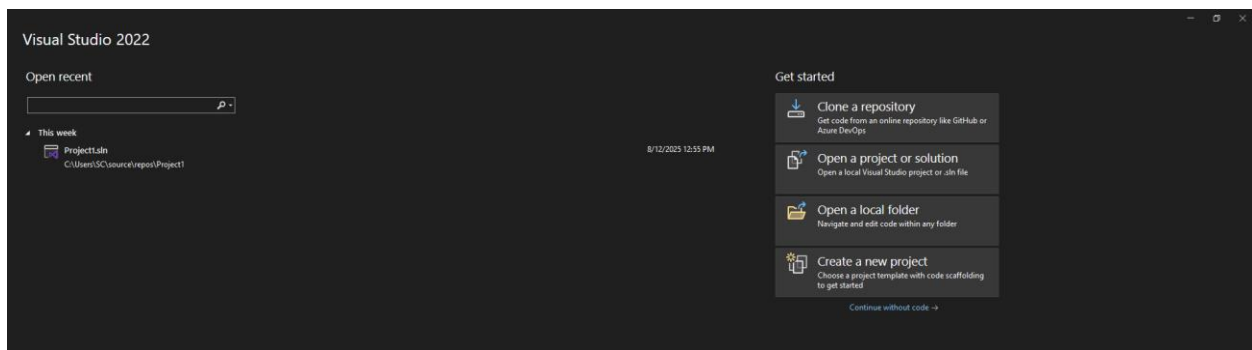


Figure 4: Creating a New Project

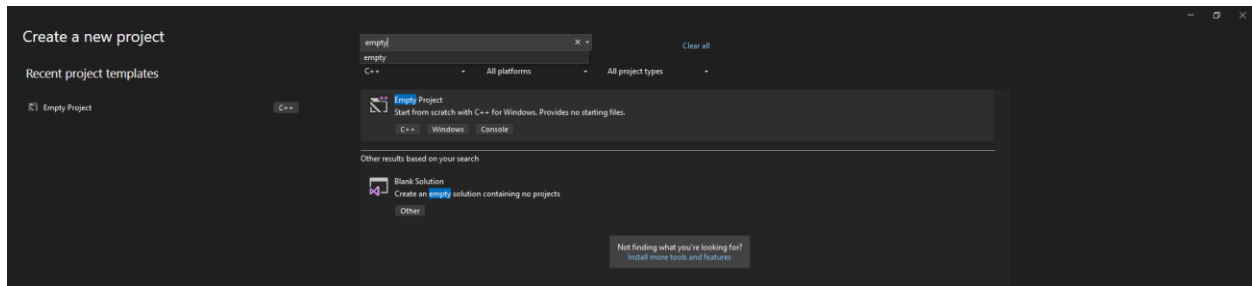


Figure 5: Template Search Bar

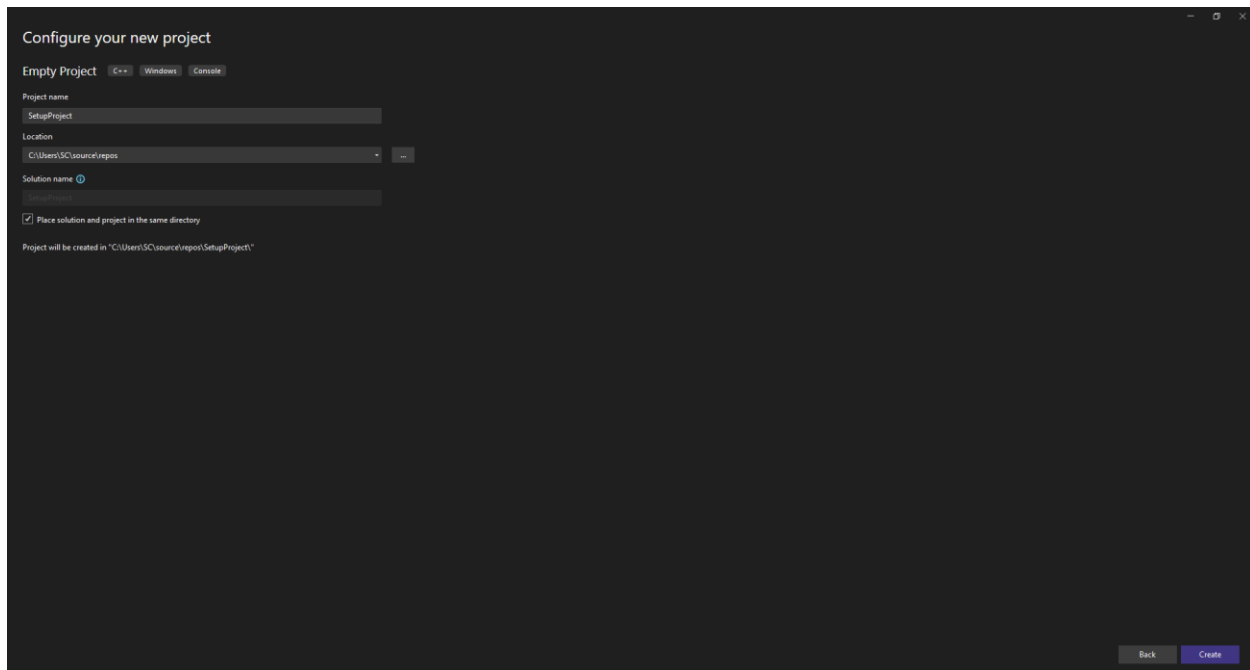


Figure 6: Project Name

Step 2: Add an Assembly File

1. In **Solution Explorer**, right-click on **Source Files** → **Add** → **New Item**.
2. Choose **Text File**, name it with a .asm extension (e.g., program.asm), and click **Add**.

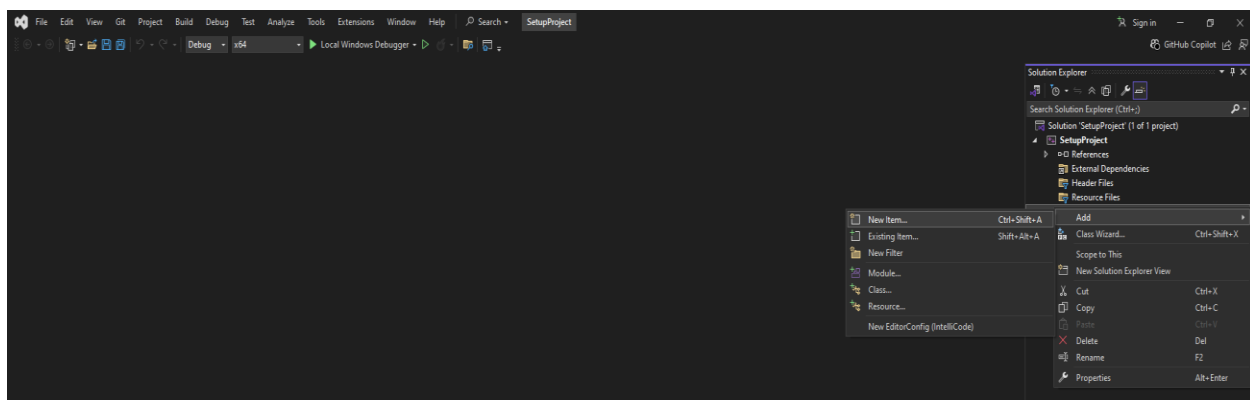


Figure 7: Inserting New Item

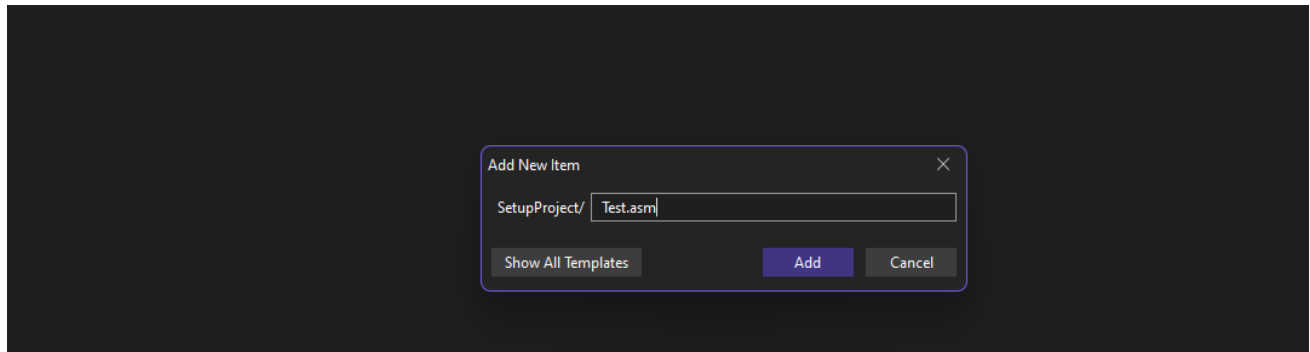


Figure 8: Renaming a File Name with .asm extension

Step 3: Enable MASM (Microsoft Macro Assembler)

1. In **Solution Explorer**, right-click on your **Project Name**.
2. Select **Build Dependencies** → **Build Customizations**.
3. Tick the checkbox for **masm (.targets, .props)** and click **OK**.

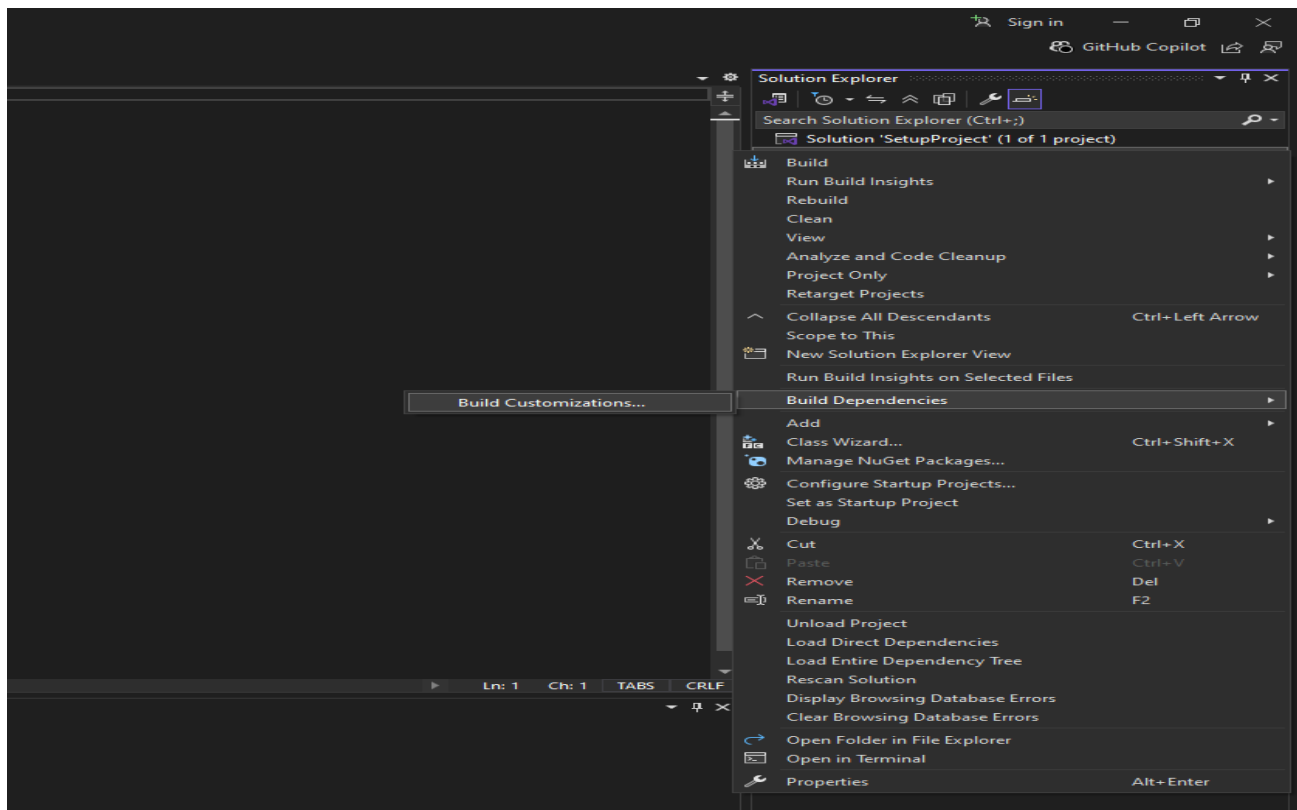


Figure 9: Build Dependencies

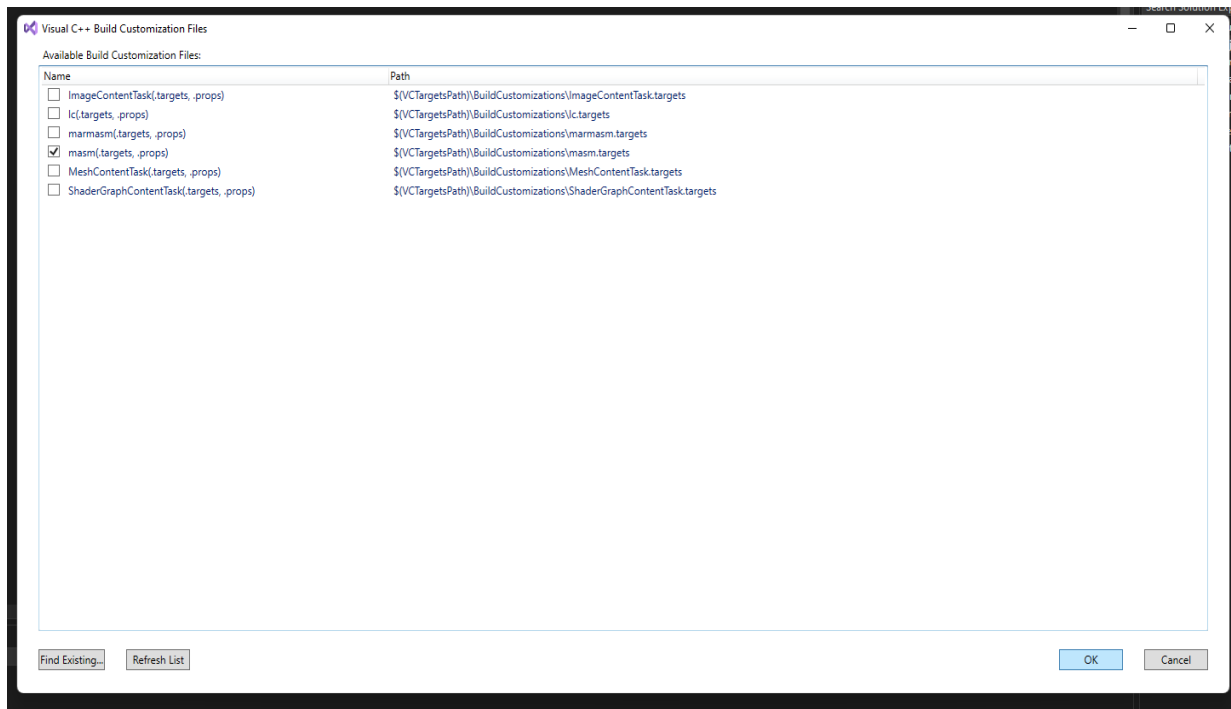


Figure 10: Tick MASM

Step 4: Configure Library Path

1. Copy the path where the **Irvine Library** is installed (e.g., C:\Irvine).
2. Right-click on your **Project Name** → select **Properties**.
3. Go to **Linker** → **General** → **Additional Library Directories**.
4. Click **Edit**, paste the copied Irvine Library path, and click **OK**.

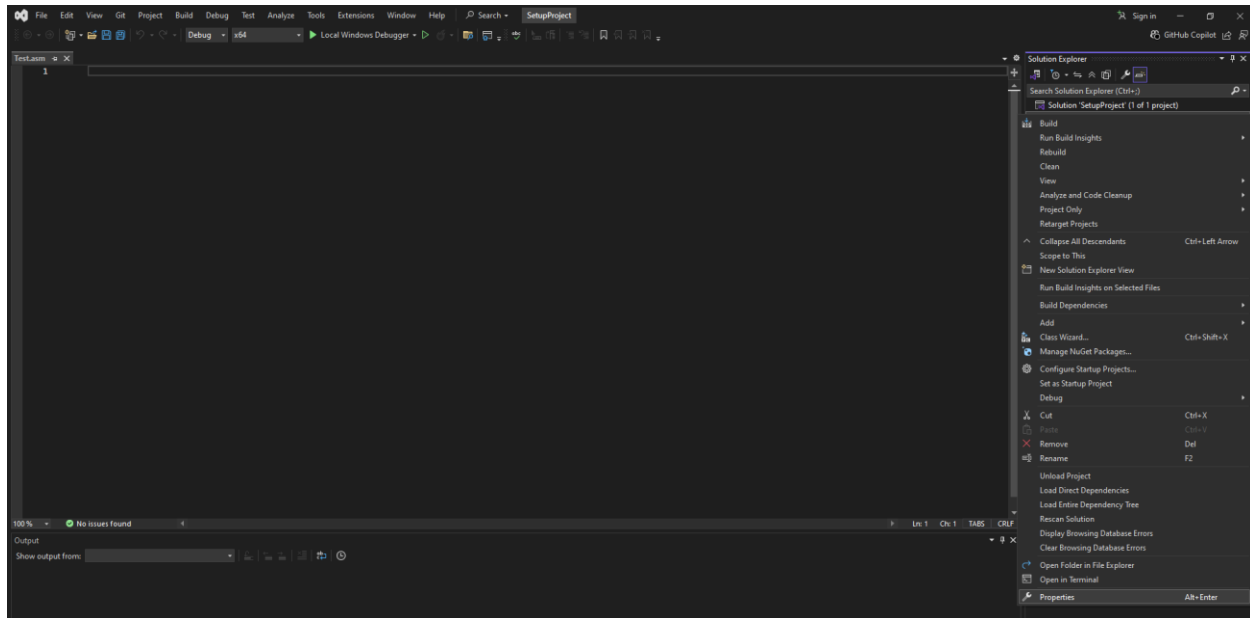


Figure 11: Go to Properties

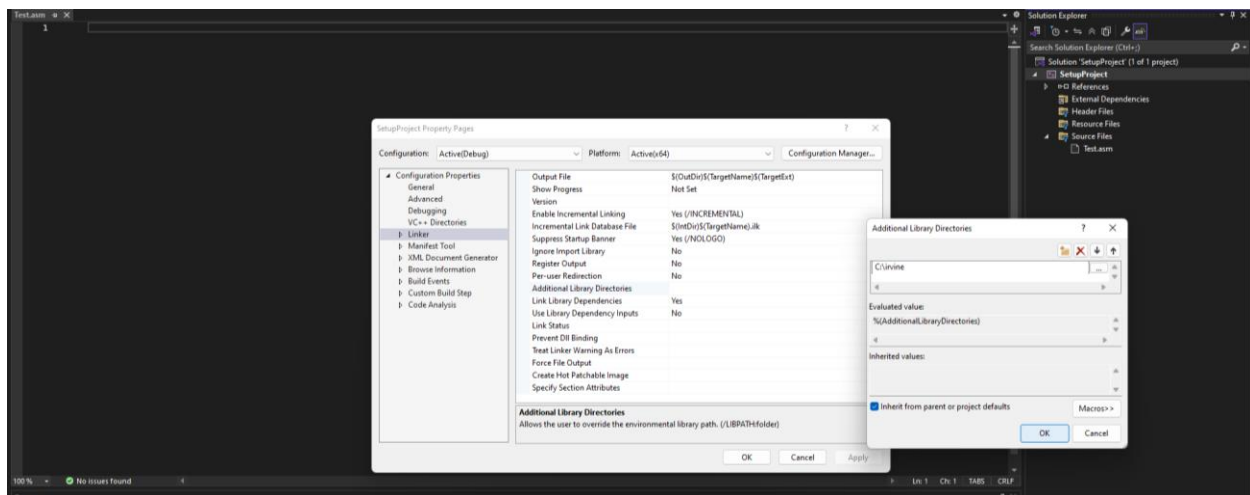


Figure 12: Copying Irvine Library Path

Step 5: Configure Linker Input

1. In the same Properties window, go to **Linker** → **Input**.
2. In **Additional Dependencies**, add:
3. Irvine32.lib
4. Click **Apply** and then **OK**.

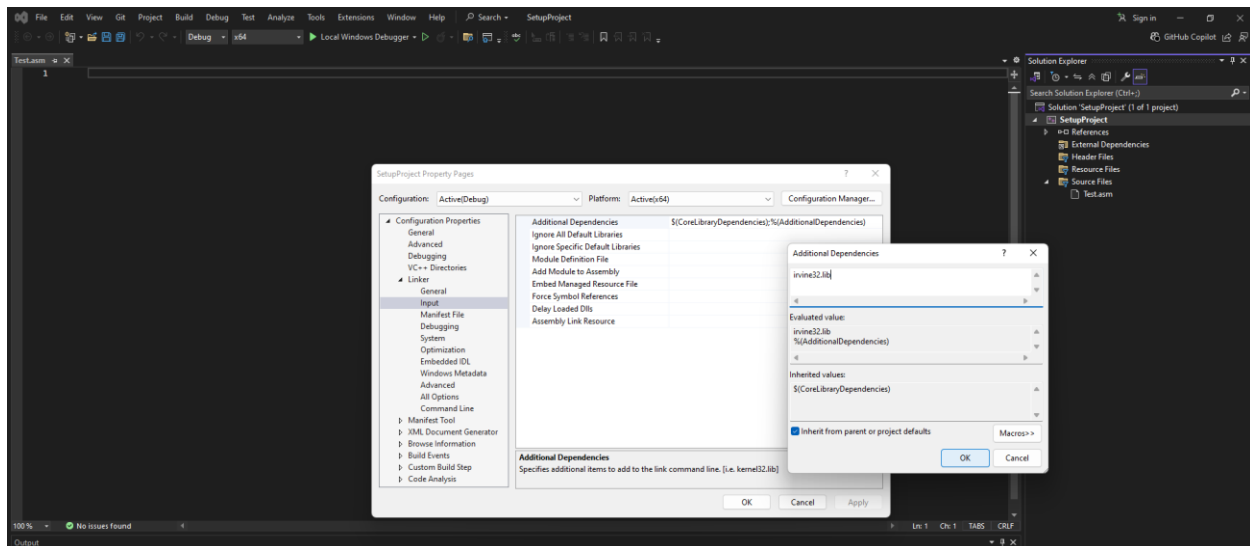


Figure 13: Configure Linker Input

Step 6: Configure ASM File Properties

1. Right-click your `.asm` file in **Solution Explorer** → **Properties**.
2. Under **General**, set **Item Type** to **Microsoft Macro Assembler**.
3. Apply the changes.

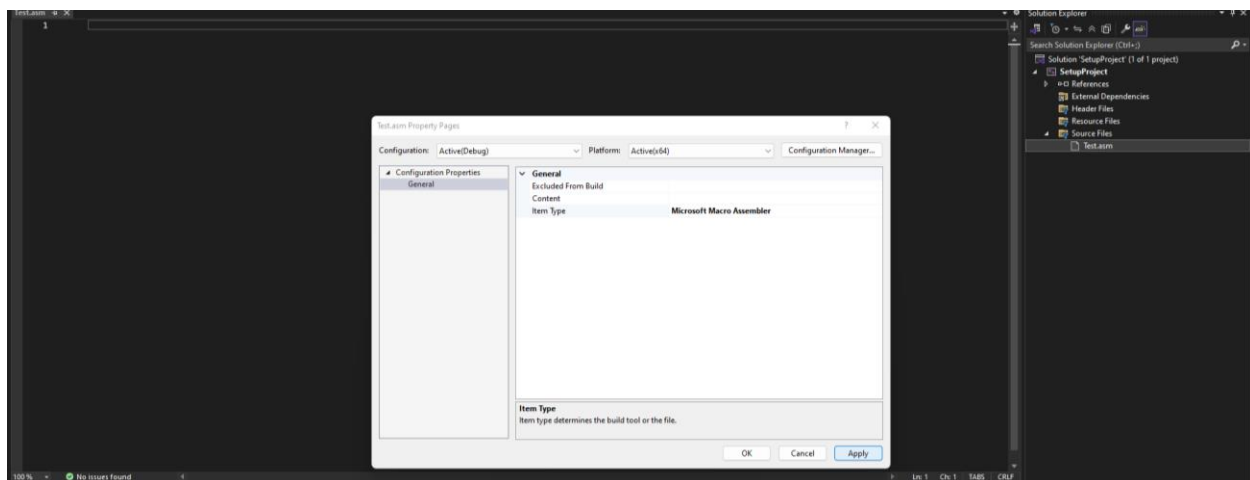


Figure 14: Configure ASM File Properties

Step 7: Set Include Path

1. In the same properties window, go to **Microsoft Macro Assembler** → **General**.
2. In **Include Paths**, paste the path of the Irvine library (e.g., C:\Irvine).
3. Click **Apply** and then **OK**.

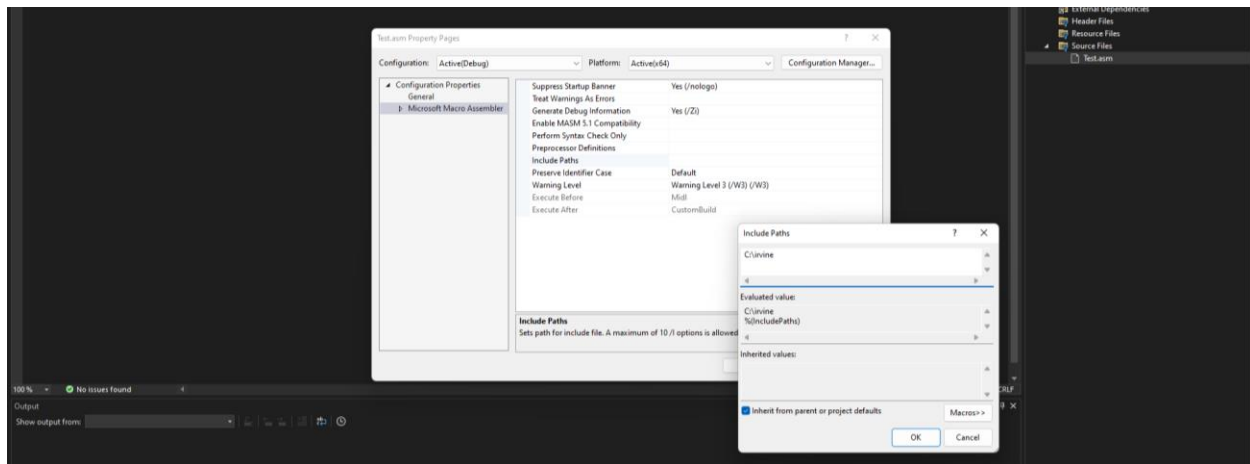


Figure 15: Set Include Path

Step 8: Write and Run a Sample Program

1. Copy a sample Irvine program (e.g., Hello World).
2. Paste it inside your `.asm` file.
3. Press **F5** to build and run the program in the debugger.

Example – 01

```
TITLE MyFirstProgram (Test.asm)

INCLUDE Irvine32.inc

.code

main PROC

    mov eax, 10h

    mov ebx, 25h

    call DumpRegs

    exit

main ENDP

END main
```

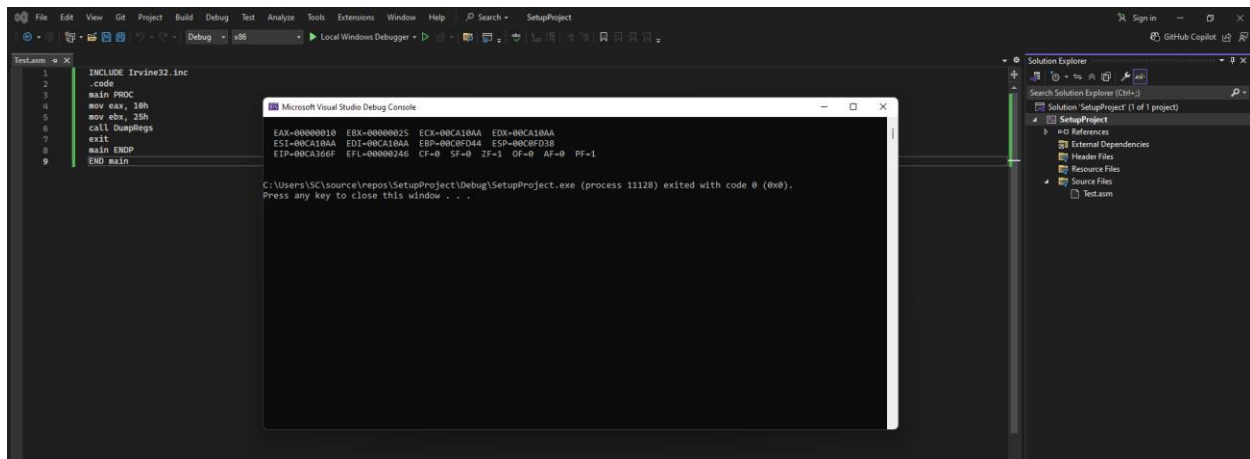


Figure 16: Example 01

Example 01 – Program Explanation

1. The first line `TITLE MyFirstProgram (Test.asm)` gives an optional title to our program.
2. The second line `INCLUDE irvine32.inc` adds a reference to the include file that links your program to the Irvine library.
3. The third line `.code` defines the beginning of the code segment (to be covered in detail later). The code segment is the segment of memory where all your code resides.
4. In the fourth line, a main procedure is defined.
5. The fifth and sixth lines show a mnemonic `mov` (to be covered in detail later) that ‘moves’ values `10h` and `25h` to `eax` and `ebx`, respectively. The radix `h` defines a hexadecimal constant. The lines seven and eight call the procedure `DumpRegs` that outputs the current values of the registers followed by a call to windows procedure named `exit` that halts the program.
6. The lines nine and ten mark the end of the main procedure.

Important Note:

The [Irvine Library](#) works only with **x86 (32-bit)** projects, not **x64**.

To check this setting:

1. Go to [Build → Configuration Manager](#).
2. Under [Active Solution Platform](#), if it shows **x64**, change it to **x86**.

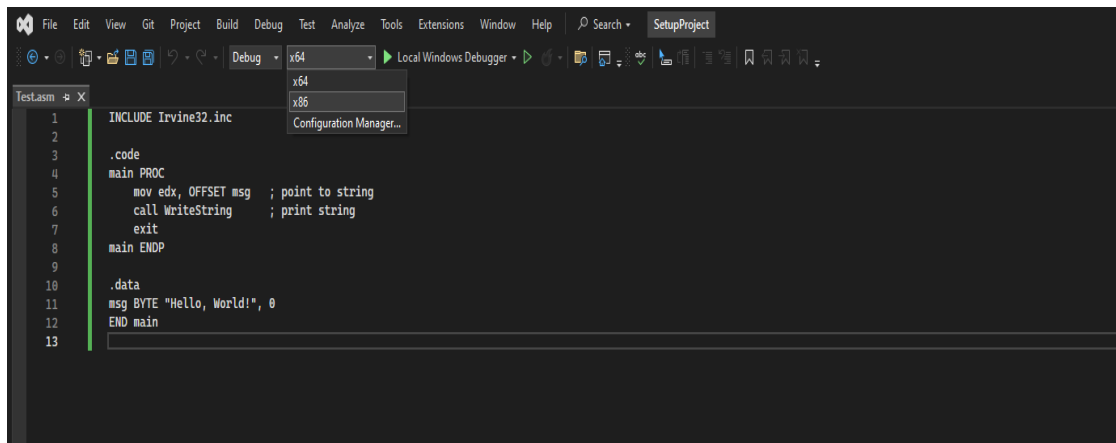


Figure 17: Configuration Manager Settings

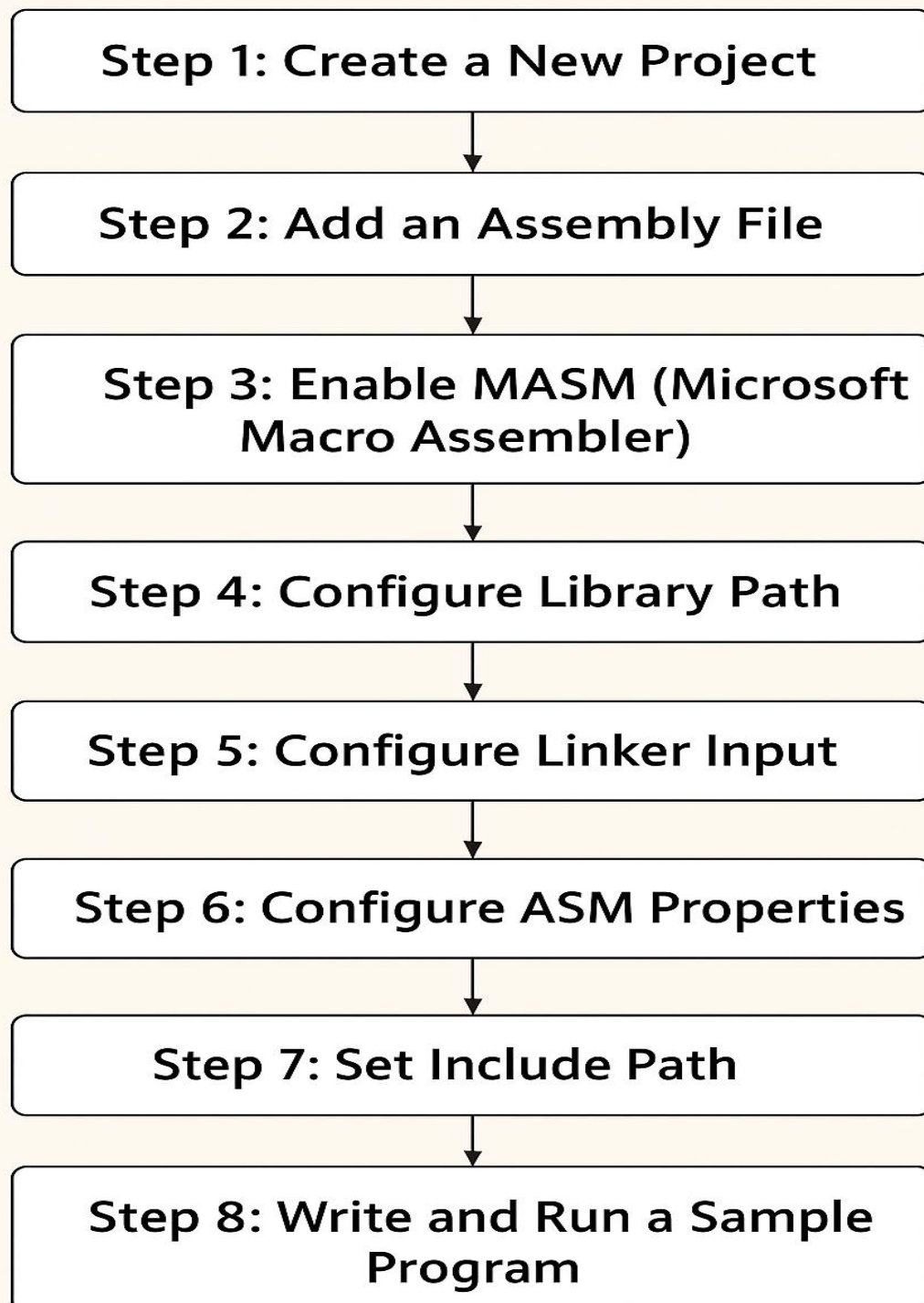


Figure 18: Assembly Project Setup Steps

Debugging our Program

We have already seen how to configure [Visual Studio 2022](#) for Assembly Language and tested it with a sample program. The output of our sample program appeared in the console window. However, in most cases, it is more useful to [watch the step-by-step execution of our program](#) by using [breakpoints](#).

In this section, we will first define important debugging terms in Visual Studio, and then cover an example for better understanding.

Debugger

The [Visual Studio Debugger](#) allows us to observe the [run-time behavior](#) of our program and locate problems. Using the debugger, we can:

- Pause execution at specific points.
- Inspect and edit variables.
- View processor registers.
- Analyze instructions generated from our source code.
- Examine the memory space used by our application.

Breakpoint

A [breakpoint](#) is a marker that tells the debugger to temporarily [suspend execution](#) of your program at a specified line of code.

- When the program halts at a breakpoint, it enters [break mode](#), allowing you to carefully examine the state of your program.

Code Stepping

[Stepping](#) is one of the most common debugging techniques. It allows you to [execute code one line at a time](#), making it easier to understand program flow and catch logical errors.

Visual Studio provides three main stepping commands under the [Debug menu](#):

- [Step Into](#) → Executes the current line of code and enters any function calls inside it. (*Shortcut: F11*)
- [Step Over](#) → Executes the current line but skips over function calls. (*Shortcut: F10*)

- **Step Out** → Executes the rest of the current function and returns to the caller. (Shortcut: **Shift + F11**)

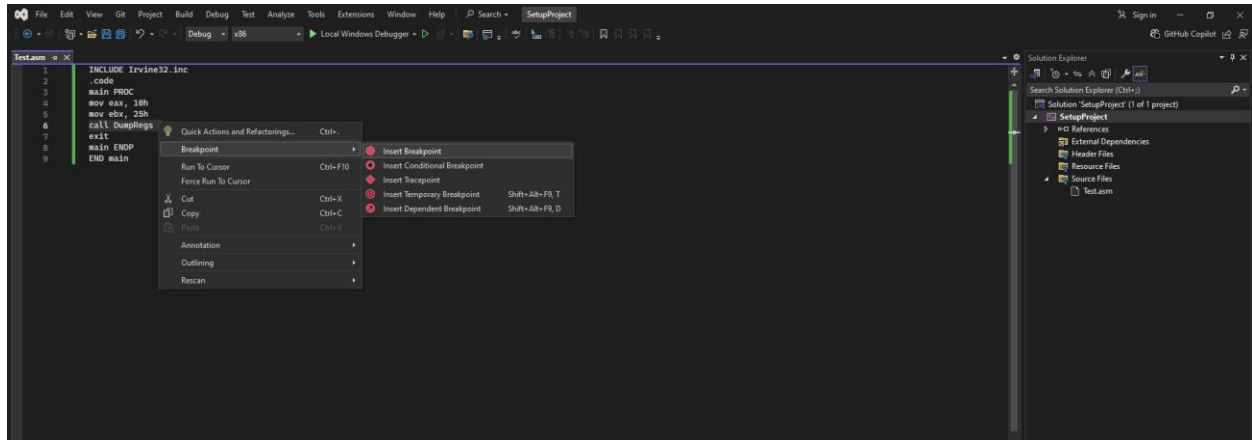


Figure 19: Break Points

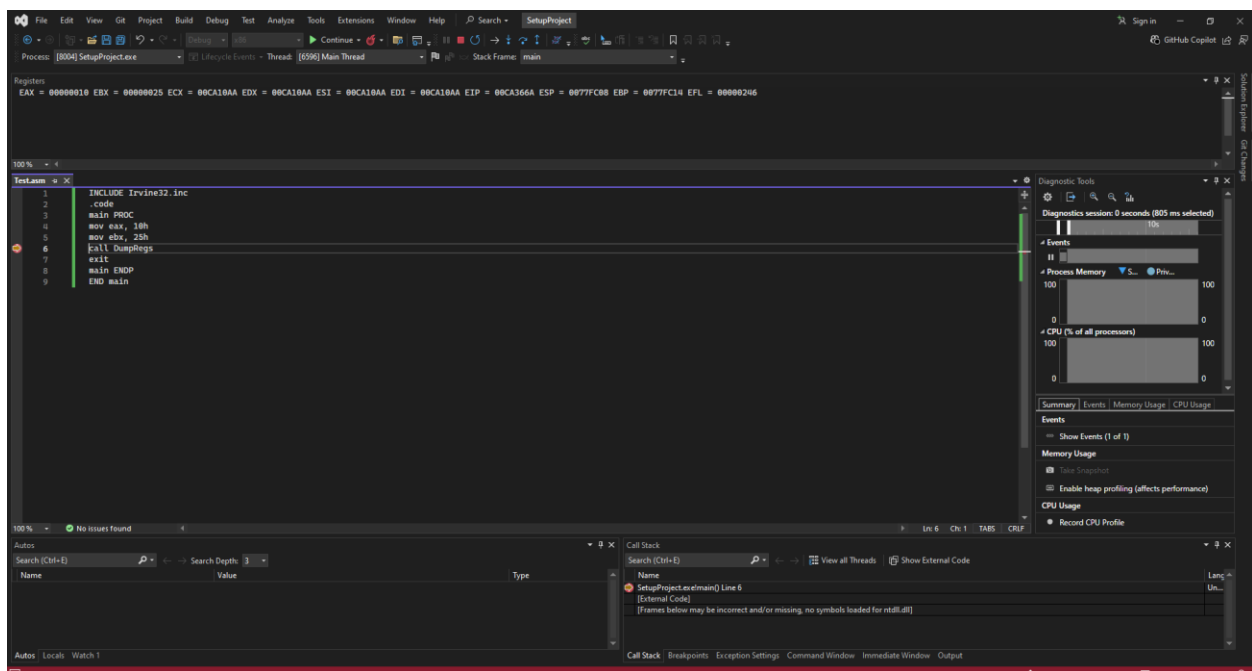


Figure 20: Break Points - Observing Register Values

Exercise

Task # 01:

Use the lab manual instructions to install Visual Studio Community Edition and set up a new Empty Project in Visual C++.

Task # 02:

- Copy the code written below and paste into the project you created by following instructions mentioned in the lab manual.
- Modify the program by replacing the 'Hello World!' message with your own Name and ID, and then execute the code.

```
INCLUDE Irvine32.inc
.code
main PROC
    mov edx, OFFSET msg
    call WriteString
    exit
main ENDP
.data
msg BYTE "Hello, World!", 0
END main
```

WHAT YOU NEED TO SUBMIT

A screenshot of the console window Displaying your Full Name and ID